Project Executable Files

Date	28 June 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	No
6	Deployed application URL (if hosted)	NA
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Credit Card Approval Prediction/

```
├── app.py
├── train.py
├── random_forest_model.pkl
├── clean_dataset.csv
├── requirements.txt
├── README.md
├── .gitignore
│
├── templates/
│   ├── index.html
│   └── result.html
│
├── static/
│   ├── style.css
│   └── images/
│
├── screenshots/
│   ├── Home_Page.png
│   ├── Approved_Result.png
│   └── Rejected_Result.png
│
├── test_cases/
│   └── Test_Results.pdf
│
└── documentation/
    ├── Project_Report.pdf
    └── Demo_Video_Link.txt
```

Step 3: Deployment / Access Details

Field	Details
Deployed URL	N/A
Login Credentials (Demo)	243ta04037@recw.ac.in 243TA04037
Platform / Hosting Provider	Local System (Flask)
Repository Link	https://github.com/243ta04037-induja/Credit-Card-Approval-Predicition/new/main
Demo Video Link	https://drive.google.com/file/d/1rd8mP0BBYL9UmlqOc_1YTfkfZ32oMFhl/view?usp=sharing

Step 4: Run Instructions

1. Install **Python 3.x** on your system.
2. Open the **Credit Card Approval Prediction** project folder in **Visual Studio Code**.
3. Install the required Python libraries using:

pip install -r requirements.txt

4. Run the Flask application using:

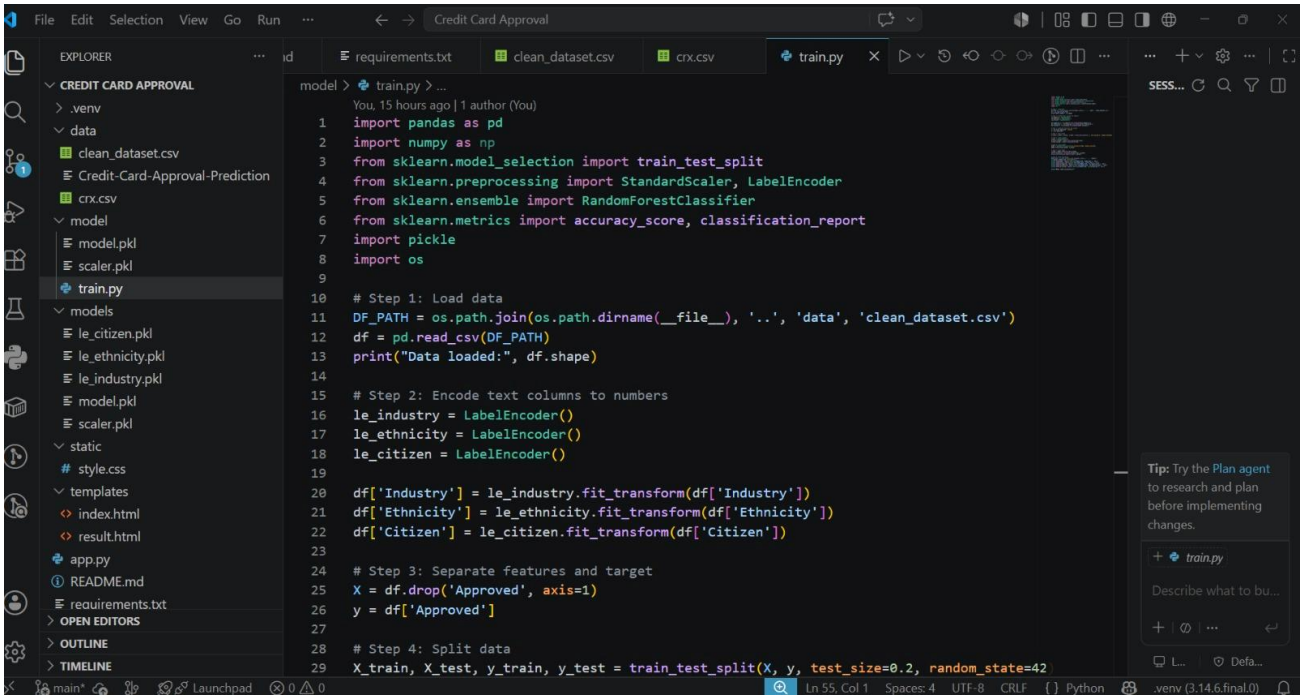
python app.py

5. Open the URL displayed in the terminal (usually **http://127.0.0.1:5000**) in your web browser.
6. Enter the required applicant details, such as age, income, employment status, credit score, and other information.
7. Click the **"Predict Approval"** button.
8. The system will process the input using the trained **Random Forest** machine learning model and display the prediction as **Approved** or **Rejected**, along with the confidence score.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction accuracy depends on the quality and size of the training dataset.	Use a larger, more diverse, and better-quality dataset to improve accuracy.
2	The application currently runs only on the local system using the Flask development server.	Deploy the application on Render, PythonAnywhere, or another cloud platform.
3	Applicant details are not stored in a database.	Integrate a database such as MySQL or SQLite in future versions.
4	The system does not provide user login or authentication.	Add user authentication and authorization in future updates.
5	The model currently uses only the Random Forest algorithm.	Compare and optimize performance using algorithms such as XGBoost, Gradient Boosting, or Neural Networks.

Code Implementation Screenshots


File Edit Selection View Go Run
Credit Card Approval
EXPLORER
requirements.txt
clean_dataset.csv
crx.csv
train.py
CREDIT CARD APPROVAL
.venv
data
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static
style.css
templates
index.html
result.html
app.py
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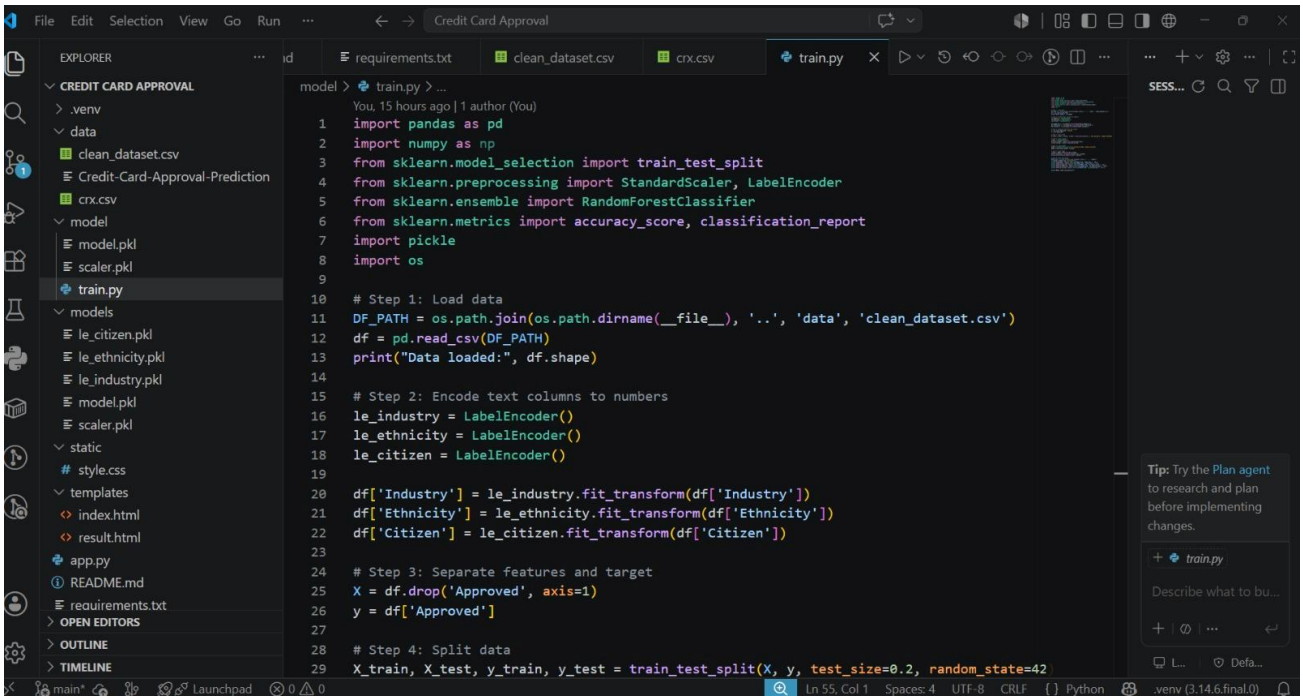
OPEN EDITORS
OUTLINE
TIMELINE
model > train.py > ...
You, 15 hours ago | 1 author (You)
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
import pickle
import os
# Step 1: Load data
DF_PATH = os.path.join(os.path.dirname(__file__), '..', 'data', 'clean_dataset.csv')
df = pd.read_csv(DF_PATH)
print("Data loaded:", df.shape)
# Step 2: Encode text columns to numbers
le_industry = LabelEncoder()
le_ethnicity = LabelEncoder()
le_citizen = LabelEncoder()
df['Industry'] = le_industry.fit_transform(df['Industry'])
df['Ethnicity'] = le_ethnicity.fit_transform(df['Ethnicity'])
df['Citizen'] = le_citizen.fit_transform(df['Citizen'])
# Step 3: Separate features and target
X = df.drop('Approved', axis=1)
y = df['Approved']
# Step 4: Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42
Tip: Try the Plan agent to research and plan before implementing changes.
train.py
Describe what to bu...
main* Launchpad
Ln 55, Col 1 Spaces: 4 UTF-8 CRLF Python .venv (3.14.6.final.0)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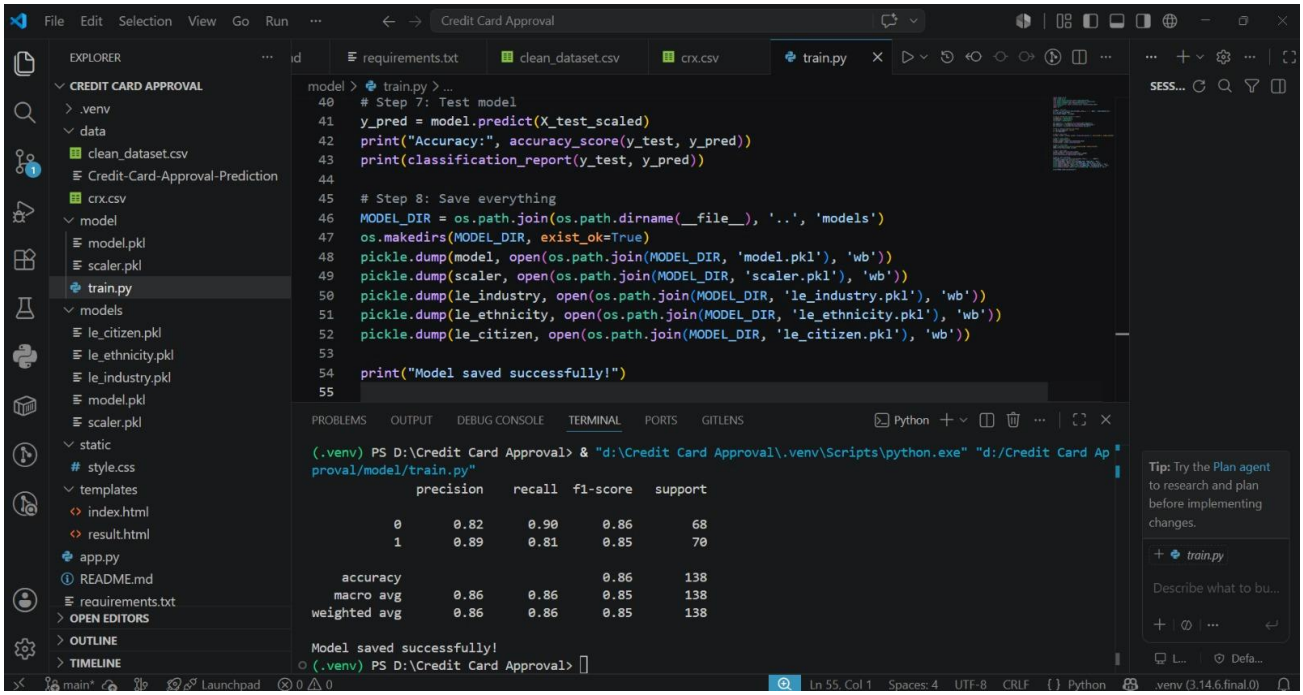
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model > train.py > ...
# Step 7: Test model
y_pred = model.predict(X_test_scaled)
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
# Step 8: Save everything
MODEL_DIR = os.path.join(os.path.dirname(__file__), '..', 'models')
os.makedirs(MODEL_DIR, exist_ok=True)
pickle.dump(model, open(os.path.join(MODEL_DIR, 'model.pkl'), 'wb'))
pickle.dump(scaler, open(os.path.join(MODEL_DIR, 'scaler.pkl'), 'wb'))
pickle.dump(le_industry, open(os.path.join(MODEL_DIR, 'le_industry.pkl'), 'wb'))
pickle.dump(le_ethnicity, open(os.path.join(MODEL_DIR, 'le_ethnicity.pkl'), 'wb'))
pickle.dump(le_citizen, open(os.path.join(MODEL_DIR, 'le_citizen.pkl'), 'wb'))
print("Model saved successfully!")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
Python
(.venv) PS D:\Credit Card Approval> & "d:\Credit Card Approval\.venv\Scripts\python.exe" "d:/Credit Card Approval/model/train.py"
precision recall f1-score support
0 0.82 0.90 0.86 68
1 0.89 0.81 0.85 70
accuracy 0.86 138
macro avg 0.86 0.86 0.85 138
weighted avg 0.86 0.86 0.85 138
Model saved successfully!
(.venv) PS D:\Credit Card Approval>
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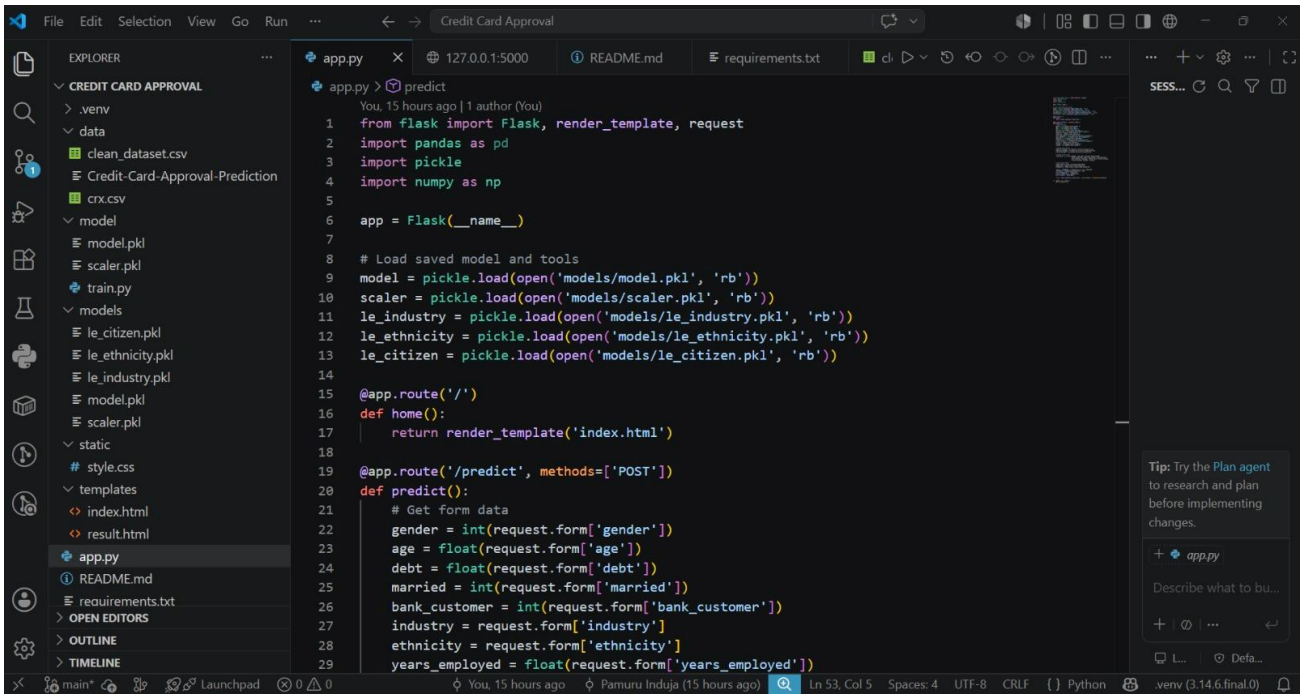
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OPEN EDITORS
OUTLINE
TIMELINE
app.py
127.0.0.1:5000
README.md
requirements.txt
app.py > predict
You, 15 hours ago | 1 author (You)
from flask import Flask, render_template, request
import pandas as pd
import pickle
import numpy as np
app = Flask(__name__)
# Load saved model and tools
model = pickle.load(open('models/model.pkl', 'rb'))
scaler = pickle.load(open('models/scaler.pkl', 'rb'))
le_industry = pickle.load(open('models/le_industry.pkl', 'rb'))
le_ethnicity = pickle.load(open('models/le_ethnicity.pkl', 'rb'))
le_citizen = pickle.load(open('models/le_citizen.pkl', 'rb'))
@app.route('/')
def home():
return render_template('index.html')
@app.route('/predict', methods=['POST'])
def predict():
# Get form data
gender = int(request.form['gender'])
age = float(request.form['age'])
debt = float(request.form['debt'])
married = int(request.form['married'])
bank_customer = int(request.form['bank_customer'])
industry = request.form['industry']
ethnicity = request.form['ethnicity']
years_employed = float(request.form['years_employed'])
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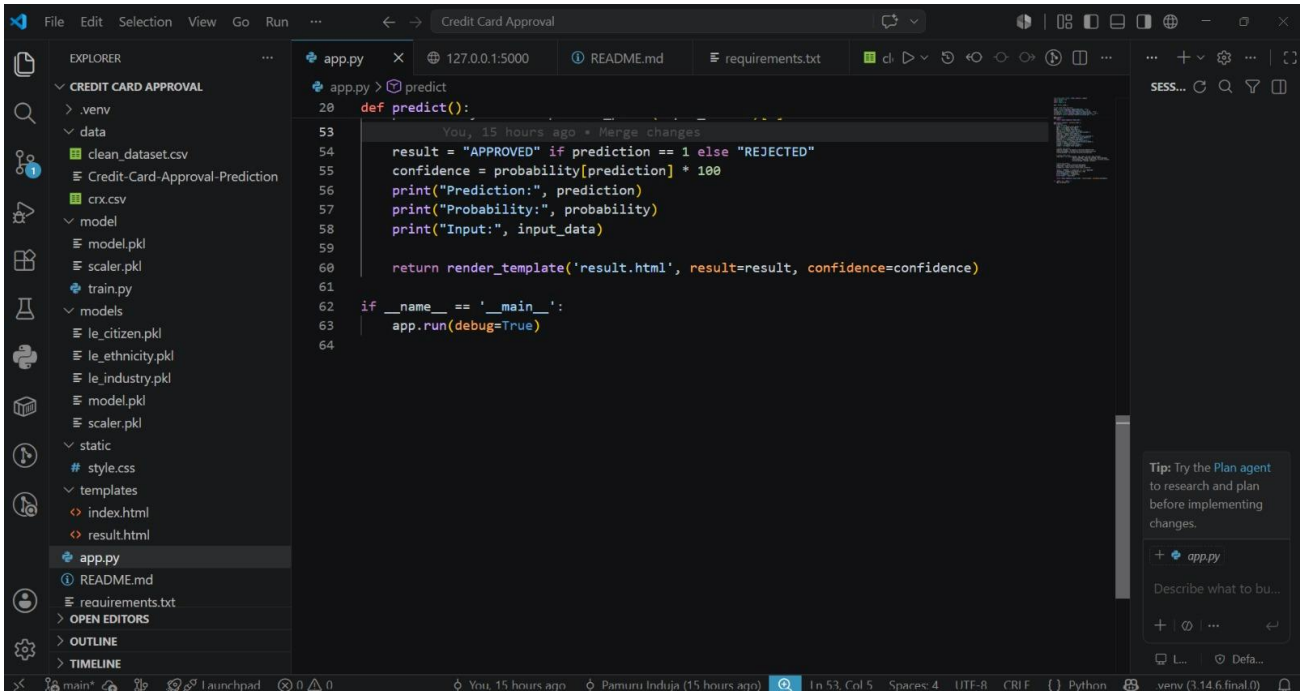
app.py
Describe what to bu...
main* Launchpad 0 0 You, 15 hours ago Pamuru Induja (15 hours ago) Ln 53, Col 5 Spaces: 4 UTF-8 CRLF Python .venv (3.14.6.final.0)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app.py > predict
def predict():
You, 15 hours ago • Merge changes
result = "APPROVED" if prediction == 1 else "REJECTED"
confidence = probability[prediction] * 100
print("Prediction:", prediction)
print("Probability:", probability)
print("Input:", input_data)
return render_template('result.html', result=result, confidence=confidence)
if __name__ == '__main__':
app.run(debug=True)
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Describe what to bu...
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You, 15 hours ago
Pamuru Induja (15 hours ago)
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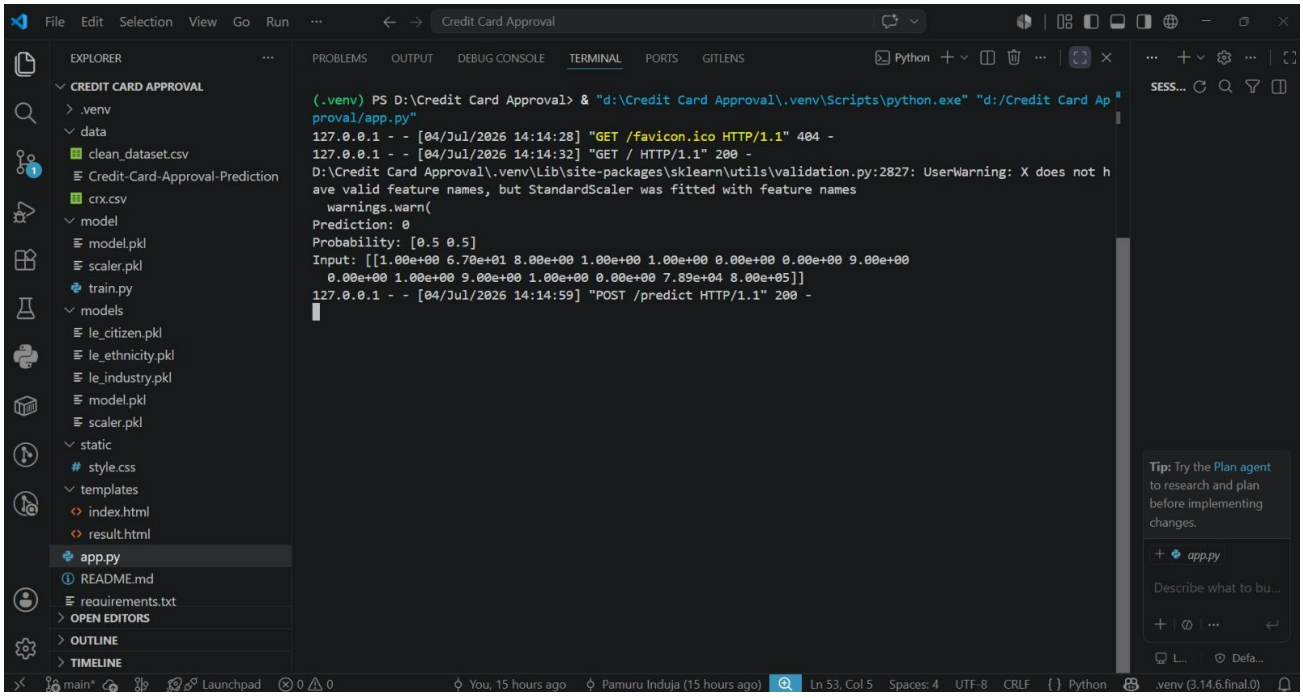
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Python
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127.0.0.1 - - [04/Jul/2026 14:14:28] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [04/Jul/2026 14:14:32] "GET / HTTP/1.1" 200 -
D:\Credit Card Approval\.venv\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
Prediction: 0
Probability: [0.5 0.5]
Input: [[1.00e+00 6.70e+01 8.00e+00 1.00e+00 1.00e+00 0.00e+00 0.00e+00 9.00e+00
0.00e+00 1.00e+00 9.00e+00 1.00e+00 0.00e+00 7.89e+04 8.00e+05]]
127.0.0.1 - - [04/Jul/2026 14:14:59] "POST /predict HTTP/1.1" 200 -
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You, 15 hours ago
Pamuru Induja (15 hours ago)
Ln 53, Col 5 Spaces: 4 UTF-8 CRLF Python
venv (3.14.6.final.0)